Mantavya Maan

B.S. Mathematics | Indian Institute of Technology Bombay
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Education

Indian Institute of Technology Bombay CPI: **8.86/10**
B.S. in Mathematics; Minor in Game Theory and Logic, Department of Computer Science and Engineering *Expected May 2027*

Scholastic Achievements

- Top **0.49** percentile in **JEE Main 2023** (1.1M candidates) and top **0.97** percentile in **JEE Advanced 2023** (0.18M candidates).

Professional Experience

Palpx | *Summer Intern, Mathematical Modeling & Artificial Intelligence* May 2026 – Jul 2026

- Architected an **LLM routing control plane** across **100+ models** using Bayesian quality estimation and utility scoring.
- Formulated **4 decision-intelligence scoring models** for lead, employee, project, and growth evaluation using statistical inference.
- Built a **multi-agent runtime** and a **Retrieval-Augmented Generation (RAG)** pipeline with source grounding and cited answers.

Gradpipe | *Machine Learning & Data Engineering Intern* Nov 2025 – Dec 2025

- Built a **talent discovery engine** on **vector databases**, combining embeddings and sparse retrieval for hybrid **semantic search**.
- Engineered an **AST-based static analysis pipeline** scoring repositories on cyclomatic complexity, concurrency, and memory use.
- Designed an **LLM-as-a-Judge** evaluation framework delivering reproducible benchmark scoring across candidate model outputs.

Fino Payments Bank | *Summer Intern, Data Analytics* Jun 2025 – Jul 2025

- Analyzed **1M+ financial records** using **SQL**, **Pandas**, and **NumPy** to identify market trends and evolving customer behavior.
- Developed interactive **Tableau dashboards** tracking **20+ KPIs** to accelerate executive decision-making and business reporting.
- Partnered with cross-functional teams to strengthen **data integrity**, data pipelines, reporting, and business analytics processes.

Key Projects

Neural Gateway | *AI Infrastructure & Multi-LLM Orchestration* May 2026 – Jul 2026

- Built a **FastAPI** routing engine scoring models on a **7-dimension utility**, with a KNN parser trained on **6,000+** labeled prompts.
- Implemented vectorized **Pareto reduction** with a cost-tightened epsilon, running **5–10× faster** across 200+ candidate models.
- Estimated routing confidence via **1,500 Thompson sampling** simulations over Beta posteriors, **abstaining** below threshold.
- Cut end-to-end routing latency by **65%** (2,250 to 830 ms) through caching and vectorization, with zero change in routing decisions.

Prompt Optimizer | *Autonomous LLMOps Platform* May 2026 – Jun 2026

- Built an **anytime-valid evaluation harness** using confidence sequences, allowing repeated looks at results without inflating error.
- Designed a **champion–challenger gate** enforcing train-margin, format, adversarial, and p95-latency fences before promotion.
- Evolved prompts through **genetic crossover** and few-shot auto-discovery, **A/B testing** challengers on **10%** of live user traffic.

Multi-Agent Orchestrator | *AI Workflow Engine* Jun 2026 – Jul 2026

- Architected a **Map-Reduce DAG runtime** where a planner decomposes user objectives into dependency-ordered atomic subtasks.
- Engineered a scheduler dispatching ready tasks to **parallel workers** via the **LangGraph** Send API with dynamic tool binding.
- Implemented cross-session durability through async **SQLite checkpointing** and **ChromaDB**-backed episodic and semantic memory.

Lead Score Model | *Mathematical Modeling and Decision Intelligence* May 2026 – Jun 2026

- Built a lead qualification framework scoring **10 parameters** across **125+ CRM variables**, from financial fit to competitive risk.
- Applied **Bayesian priors**, **Beta confidence intervals**, **sigmoid** transforms, and **exponential decay** for explainable ranking.
- Designed hard **qualification gates** and score modifiers into an **explainable 0–100 index**, deployed as an interactive console.

Technical Skills

Languages & Tools: Python, SQL, C++, C, Java, JavaScript, LaTeX, Git, GitHub, REST APIs, SQLite, Jupyter, Tableau
AI / ML: Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), NLP, RAG, AI Agents, LLMOps
GenAI Engineering: LLM Orchestration, Prompt Engineering, Model Evaluation, Embeddings, Semantic Search, Vector Databases
Frameworks & Libraries: LangChain, LangGraph, FastAPI, Pydantic, NumPy, Pandas, SciPy, ChromaDB, Pinecone, Ollama
Mathematics: Statistics, Probability, Bayesian Inference, Optimization, A/B Testing, Thompson Sampling, Sequential Testing

Leadership & Activities

- Served as Hostel **Secretary** representing **1200+ residents**, and coordinated **15+ T20 matches** during the Aavhan sports festival.
- Served as **Services** Coordinator at Mood Indigo, **Operations** Coordinator at E-Cell, and **Infrastructure** Coordinator at Techfest.
- Won **silver** in **Powerlifting** at the General Sports Championship, IITB, and **bronze** at the Institute Powerlifting Open.
- Completed Summer of Science in **Data Structures & Algorithms** and Season of Code in **Competitive Programming**.